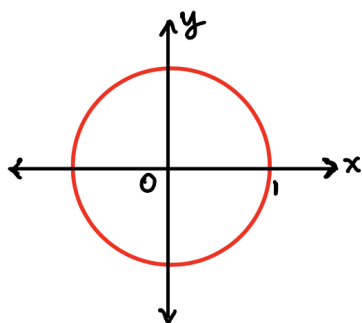


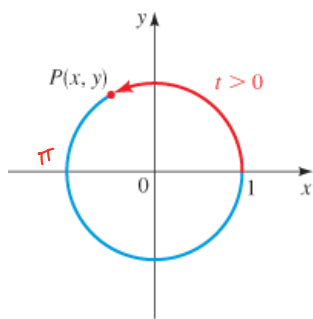
I. The Unit Circle

The unit circle is the circle of radius 1 centered at the origin in the xy-plane. Its equation is $x^2 + y^2 = 1$.

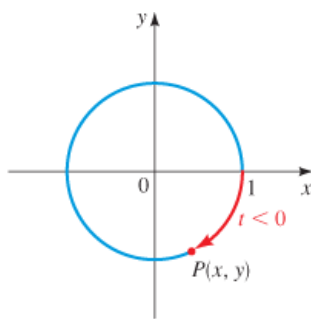
**II. Terminal Points on the Unit Circle**

Suppose t is a real number. If $t \geq 0$, let's mark off a distance t along the unit circle, starting at (1,0) and moving in a counterclockwise direction if t is positive, or in a clockwise direction if t is negative. In this way we arrive at a point $P(x, y)$, called the terminal point determined by the real number t .

The circumference of the unit circle is $C = 2\pi$. To move a point halfway around the unit circle, it travels a distance of π . To move a quarter of the distance around the circle, it travels a distance of $\frac{\pi}{2}$.

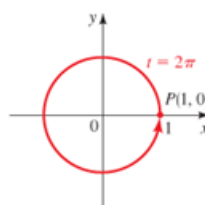
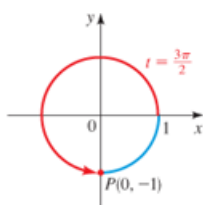
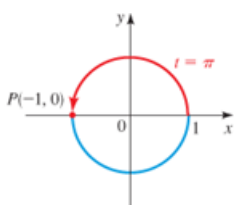
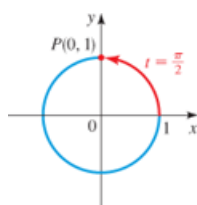


(a) Terminal point $P(x, y)$ determined by $t > 0$



(b) Terminal point $P(x, y)$ determined by $t < 0$

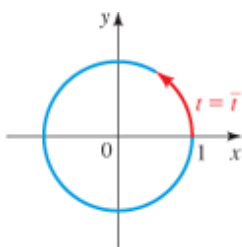
Terminal points determined by $t = \frac{\pi}{2}$, $t = \pi$, $t = \frac{3\pi}{2}$, and $t = 2\pi$.



III. The Reference Number

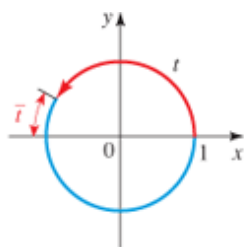
Let t be a real number. The **reference number** $\bar{t}$ associated with t is the **shortest distance along the unit circle between terminal point** determined by t and the **x-axis**. To find the reference number $\bar{t}$ it is helpful to know the quadrant in which the terminal point determined by t lies. If the terminal point lies in quadrants **I or IV**, where x is positive, we find $\bar{t}$ by moving along the circle to the **positive x-axis**. If it lies in quadrants **II or III**, where x is negative, we find $\bar{t}$ by moving along the circle to the **negative x-axis**.

Quadrant I



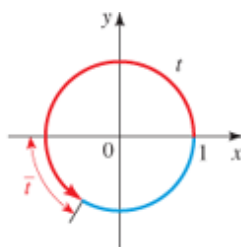
$$\bar{t} = t$$

Quadrant II



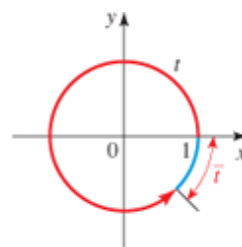
$$\bar{t} = \pi - t$$

Quadrant III



$$\bar{t} = t - \pi$$

Quadrant IV



$$\bar{t} = 2\pi - t$$

Using Reference Numbers To Find Terminal Points

Steps for finding the terminal point P determined by any value of t

1. Find the **reference number** $\bar{t}$.
2. Find the **terminal point** $Q(a, b)$ determined by $\bar{t}$.
3. The terminal point determined by t is $P(\pm a, \pm b)$, where the **signs are chosen** according to the **quadrant** in which this terminal point lies.

Since the circumference of the unit circle is 2π , the terminal point determined by t is the same as that determined by $t + 2\pi$ or $t - 2\pi$. In general, we can **add or subtract** 2π any number of times without changing the terminal point determined by t .

Example 1: Show that the point is on the unit circle.

$$\begin{aligned} \text{a. } & \left(-\frac{5}{7}, -\frac{2\sqrt{6}}{7}\right) \\ & \sqrt{\left(-\frac{5}{7} - 0\right)^2 + \left(-\frac{2\sqrt{6}}{7} - 0\right)^2} \\ &= \sqrt{\frac{25}{49} + \frac{24}{49}} \\ &= \sqrt{1} = 1 \rightarrow x^2 + y^2 = 1 \checkmark \end{aligned}$$

$$\begin{aligned} \text{b. } & \left(\frac{3}{4}, \frac{-\sqrt{7}}{4}\right) \\ & \sqrt{\left(\frac{3}{4} - 0\right)^2 + \left(\frac{-\sqrt{7}}{4} - 0\right)^2} \\ &= \sqrt{\frac{9}{16} + \frac{7}{16}} \\ &= \sqrt{1} = 1 \rightarrow x^2 + y^2 = 1 \checkmark \end{aligned}$$

Example 2: Find the missing coordinate of P , using the fact that P lies on the unit circle in the given quadrant.

Coordinates

Quadrant

$$\text{a. } P\left(-\frac{3}{5}, \quad\right)$$

III

$$\sqrt{\left(-\frac{3}{5} - 0\right)^2 + y^2} = 1$$

$$\frac{9}{25} + y^2 = 1$$

$$y = \sqrt{\frac{16}{25}} = -\frac{4}{5}$$

Coordinates

Quadrant

$$\text{b. } P\left(\quad, \frac{-2}{7}\right)$$

IV

$$\sqrt{x^2 + \left(\frac{-2}{7}\right)^2} = 1$$

$$x^2 + \frac{4}{49} = 1$$

$$x^2 = \frac{45}{49}$$

$$\begin{aligned} x &= \sqrt{\frac{45}{49}} \\ &= \frac{3\sqrt{5}}{7} \end{aligned}$$

Example 3: The point P is on the unit circle. Find $P(x, y)$ from the given information.

- a. The y -coordinate of P is $-\frac{3}{5}$, and the x -coordinate is positive.

$$\sqrt{x^2 + \left(-\frac{3}{5}\right)^2} = 1 \rightarrow x = \frac{4}{5}$$

$$x^2 + \frac{9}{25} = 1$$

$$x^2 = \frac{16}{25}$$

- b. The x -coordinate of P is positive, and the y -coordinate of P is $-\frac{\sqrt{5}}{5}$

$$\sqrt{x^2 + \left(-\frac{\sqrt{5}}{5}\right)^2} = 1$$

$$x^2 + \frac{5}{25} = 1$$

$$x = \sqrt{\frac{20}{25}}$$

$$x = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$$

- c. The x -coordinate of P is $-\frac{2}{5}$, and P lies above the x -axis. ^{positive}

$$\sqrt{\left(-\frac{2}{5}\right)^2 + y^2} = 1 \rightarrow y = \frac{\sqrt{21}}{5}$$

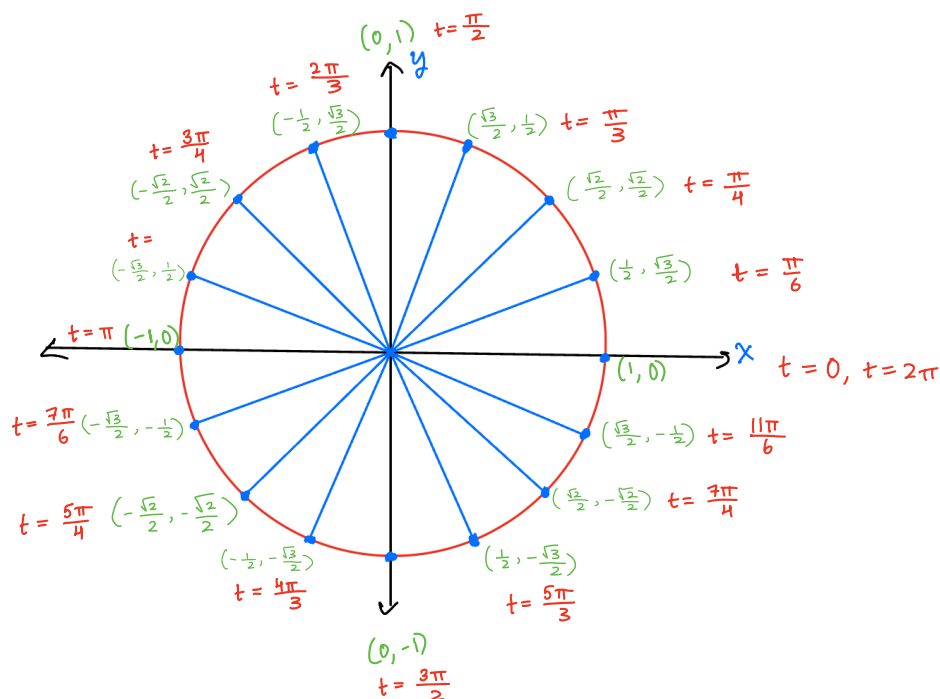
$$\frac{4}{25} + y^2 = 1$$

$$y^2 = \frac{21}{25}$$

$$y = \sqrt{\frac{21}{25}}$$

SPECIAL TERMINAL POINTS

Example 4: Find the terminal points determined by the special values of t .



Exercise 5: Find the reference number for each value of t

a. $t = 9\pi$

$\bar{t} = \pi$

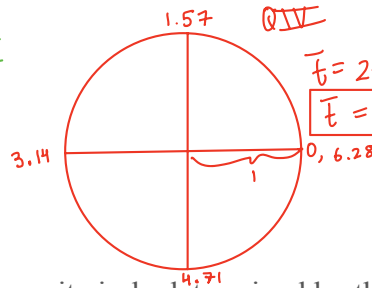
b. $t = -\frac{41\pi}{4}$

$-\frac{41\pi}{4} + \frac{48\pi}{4} = \frac{7\pi}{4}$

$2\pi - \frac{7\pi}{4}$

$\bar{t} = \frac{\pi}{4}$

c. $t = 6 \text{ rad}$



$\bar{t} = 2\pi - 6$

$\bar{t} = 0.28$

d. $t = -7$

$-7 + 12.56 = 5.56$

QIV

$2\pi - 5.56$

$6.28 - 5.56$

$\bar{t} = 0.72$

Exercise 6: Find the terminal point $P(x, y)$ on the unit circle determined by the given value of t .

a. $t = 5\pi$

$5\pi - 4\pi = \pi$

$(-1, 0)$

b. $t = -\frac{11\pi}{3}$

$= -\frac{11\pi}{3} + \frac{12\pi}{3}$

$t = \frac{\pi}{3}$

$\bar{t} = \frac{\pi}{3} \therefore \text{Q I}$

$(\frac{\sqrt{3}}{2}, \frac{1}{2})$

c. $t = \frac{5\pi}{2}$

$t = \frac{5\pi}{2} - \frac{4\pi}{2}$

$t = \frac{\pi}{2}$

$(0, 1)$

d. $t = \frac{31\pi}{6}$

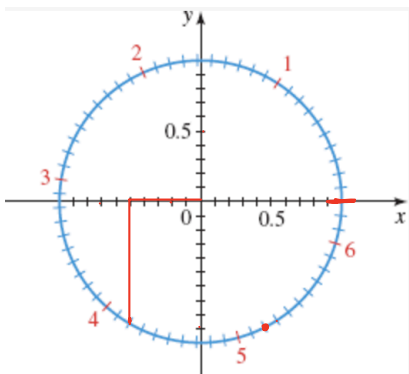
$t = \frac{31\pi}{6} - \frac{24\pi}{6}$

$t = \frac{7\pi}{6}$

$\bar{t} = \frac{7\pi}{6} - \pi \therefore \text{Q III}$

$(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$

Example 7: The unit circle is graphed in the figure below. Use the figure to find the terminal point determined by the real number t , with coordinates rounded to one decimal place.



a. $t = 2.5$

$P(-0.7, 0.6)$

b. $t = -1.1$

$P(0.4, -0.9)$

$(-0.9)^2 + x^2 = 1$

$x^2 = 0.19$

$x = \sqrt{0.19}$

x is closer to 0.4 than it is to 0.5

c. $t = 4.2$

$P(-0.5, -0.9)$

Example 8: Suppose that the terminal point determined by t is the point $(\frac{3}{5}, \frac{4}{5})$ on the unit circle. Find the terminal point determined by each of the following.

a. $-t$

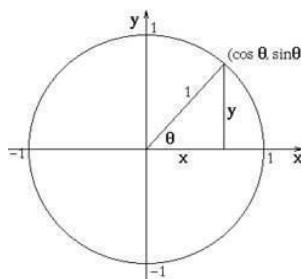
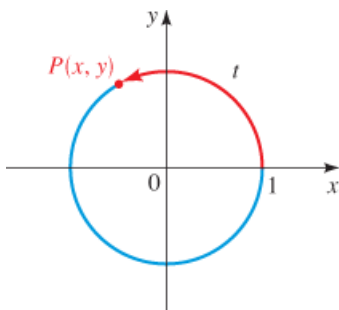
$(\frac{3}{5}, -\frac{4}{5})$

b. $\pi - t$

$(-\frac{3}{5}, \frac{4}{5})$

c. $2\pi + t$

$(\frac{3}{5}, \frac{4}{5})$

Chp 6.2 Trigonometry Functions of Real Numbers**I. The Trigonometric Functions**

The coordinates of any point on the unit circle are $(\cos t, \sin t)$

Definition of The Trigonometric Functions

Let t be any real number and let $P(x, y)$ be the terminal point on the unit circle determined by t . We define

$$\begin{array}{lll} \sin t = y & \cos t = x & \tan t = \frac{y}{x} \quad (x \neq 0) \\ \csc t = \frac{1}{y} \quad (y \neq 0) & \sec t = \frac{1}{x} \quad (x \neq 0) & \cot t = \frac{x}{y} \quad (y \neq 0) \end{array}$$

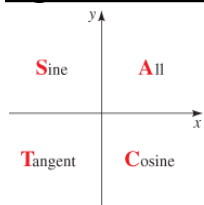
Because the trigonometric functions can be defined in terms of the unit circle, they are sometimes called the **circular functions**.

Domain of the Trigonometric Functions

Function	Domain
$\sin, \cos$	all real numbers.
$\tan, \sec$	all real numbers other than $\pi/2 + n\pi$ for any integer n
$\cot, \csc$	all real numbers other than $n\pi$ for any integer n

II. Values of the Trigonometric Functions

To compute other values of the trigonometric functions, we first determine **their signs**. The signs of the trigonometric functions depend on the quadrant in which the **terminal point** of t lies.

Signs of the Trigonometric Functions.

Quadrant	Positive Functions	Negative Functions
I	all	None
II	$\sin, \csc$	$\cos, \sec, \tan, \cot$
III	$\tan, \cot$	$\sin, \csc, \cos, \sec$
IV	$\cos, \sec$	$\sin, \csc, \tan, \cot$

Since the trigonometric functions are defined in terms of the coordinates of terminal points, we can use the **reference number** to find values of the trigonometric functions.

Evaluating Trigonometric Functions for Any Real Number

To find the values of the trigonometric functions for any real number, we carry out the following steps.

1. **Find the reference number.** Find the reference number associated with.
2. **Find the sign.** Determine the sign of the trigonometric function by noting the quadrant in which the terminal point lies.
3. **Find the value.** The value of the trigonometric function is the same, except possibly for sign, as the value of the trigonometric function of.

Reciprocal Relation

$$\csc t = \frac{1}{\sin t} \quad \sec t = \frac{1}{\cos t} \quad \cot t = \frac{1}{\tan t}$$

Even-Odd Properties

If $f(-x) = -f(x)$, odd

If $f(-x) = f(x)$, even

$$\sin(-t) = -\sin t \text{ odd} \quad \cos(-t) = \cos t \text{ even} \quad \tan(-t) = -\tan t \text{ odd}$$

$$\csc(-t) = -\csc t \text{ odd} \quad \sec(-t) = \sec t \text{ even} \quad \cot(-t) = -\cot t \text{ odd}$$

The odd trigonometric functions are sine, cosecant, tangent, and cotangent. The even trigonometric functions are cosine and secant.

III. Fundamental Identities:

Reciprocal Identities

$$\csc t = \frac{1}{\sin t} \quad \sec t = \frac{1}{\cos t} \quad \cot t = \frac{1}{\tan t} \quad \tan t = \frac{\sin t}{\cos t} \quad \cot t = \frac{\cos t}{\sin t}$$

Pythagorean Identities

$$\sin^2 t + \cos^2 t = 1 \quad \tan^2 t + 1 = \sec^2 t \quad 1 + \cot^2 t = \csc^2 t$$

Example 1: Find the exact value of the trigonometric function at the given real number.

a. $\sin\left(-\frac{5\pi}{6}\right)$

$$\sin\left(-\frac{5\pi}{6}\right) = -\sin\left(\frac{5\pi}{6}\right)$$

reference number:

$$\pi - \frac{5\pi}{6} \because \text{Q II}$$

$$= \frac{\pi}{6}$$

$$\sin\left(-\frac{5\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right)$$

$$\sin\left(-\frac{5\pi}{6}\right) = \boxed{-\frac{1}{2}}$$

b. $\tan(15\pi)$

$$\tan(15\pi)$$

$$= \tan(\pi)$$

$$= \boxed{0}$$

c. $\csc \frac{7\pi}{3}$

$$\csc\left(\frac{7\pi}{3}\right) = \frac{1}{\sin\left(\frac{7\pi}{3}\right)}$$

$$= \frac{1}{\sin\left(\frac{\pi}{3}\right)}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}}$$

$$= \frac{2}{\sqrt{3}}$$

$$\csc\left(\frac{7\pi}{3}\right) = \boxed{\frac{2\sqrt{3}}{3}}$$

d. $\sec \frac{11\pi}{6}$

$$\sec\left(\frac{11\pi}{6}\right) = \frac{1}{\cos\left(\frac{11\pi}{6}\right)}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}}$$

$$= \frac{2}{\sqrt{3}}$$

$$\sec\left(\frac{11\pi}{6}\right) = \boxed{\frac{2\sqrt{3}}{3}}$$

Example 2: The terminal point determined by a real number t is given. Find $\sin t$, $\cos t$, and $\tan t$.

a. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

$$\sin t = -\frac{4}{5}$$

$$\cos t = \frac{3}{5}$$

$$\tan t = \frac{-\frac{4}{5}}{\frac{3}{5}} = -\frac{4}{3}$$

b. $\left(\frac{-6}{7}, \frac{-\sqrt{13}}{7}\right)$

$$\sin t = -\frac{\sqrt{13}}{7}$$

$$\cos t = -\frac{6}{7}$$

$$\tan t = \frac{-\frac{\sqrt{13}}{7}}{-\frac{6}{7}} = \frac{\sqrt{13}}{6}$$

Example 3: Find the sign of the expression if the terminal point determined by t is in the given quadrant.

a. $\sin t \tan t$, Quadrant IV

$$(-)(-) = \boxed{\text{positive}}$$

b. $\frac{\tan t \sin t}{\cot t}$, Quadrant III

$$\frac{(+)(-)}{(+)} = \frac{(-)}{(+)} = (-)$$

$\boxed{\text{negative}}$

Example 4: From the information given, find the quadrant in which the terminal point determined by t lies.

a. $\underbrace{\csc t > 0}_{\text{Q I, Q II}} \text{ and } \underbrace{\sec t < 0}_{\text{Q II, Q III}}$

$\boxed{\text{Q II}}$

b. $\underbrace{\cos t > 0}_{\text{Q I, Q IV}} \text{ and } \underbrace{\cot t < 0}_{\text{Q II, Q IV}}$

$\boxed{\text{Q IV}}$

Example 5: Write the first expression in terms of the second if the terminal point determined by t is in the given quadrant.

a. $\csc t$, $\cot t$; Quadrant II

$$\cot^2 t + 1 = \csc^2 t$$

$$\boxed{\csc t = \sqrt{\cot^2 t + 1}}$$

b. $\tan^2 t$, $\sin t$; any quadrant

$$\sin^2 t + \cos^2 t = 1$$

$$\cos t = \pm \sqrt{1 - \sin^2 t}$$

$$\tan t = \frac{\sin t}{\cos t}$$

$$\tan t = \frac{\sin t}{\pm \sqrt{1 - \sin^2 t}}$$

$$\tan^2 t = \left(\frac{\sin t}{\pm \sqrt{1 - \sin^2 t}} \right)^2$$

$$\boxed{\tan^2 t = \frac{\sin^2 t}{1 - \sin^2 t}}$$

Example 6: Find the values of the trigonometric functions of t from the given information.

a. $\cos t = \frac{-7}{25}$, terminal point of t is in Quadrant III

$$\begin{aligned} \sin^2 t + \left(\frac{-7}{25}\right)^2 &= 1 \\ \sin t &= \sqrt{\frac{576}{625}} \\ \sin t &= -\frac{24}{25} \quad \because \text{Q III} \end{aligned}$$

$$\tan t = \frac{-\frac{24}{25}}{-\frac{7}{25}} = \frac{24}{7}$$

$$\sec t = -\frac{25}{7}$$

$$\csc t = -\frac{25}{24}$$

$$\cot t = \frac{7}{24}$$

b. $\tan t = -4$, $\csc t > 0$

Q II, Q IV Q I, Q II

Q II

$$\tan^2 t + 1 = \sec^2 t$$

$$(-4)^2 + 1 = \sec^2 t$$

$$\sec t = -\sqrt{17} \quad \because \text{Q II}$$

$$\cos t = -\frac{1}{\sqrt{17}}$$

$$\cos t = -\frac{\sqrt{17}}{17}$$

$$\cot t = -\frac{1}{4}$$

$$-4 = \frac{\sin t}{-\frac{1}{\sqrt{17}}}$$

$$\sin t = \frac{4}{\sqrt{17}}$$

$$\csc t = \frac{1}{\frac{4}{\sqrt{17}}}$$

$$\csc t = \frac{\sqrt{17}}{4}$$

c. $\sin t = \frac{-1}{4}$, $\sec t < 0$

Q III, Q IV

Q II, Q III

Q III

$$\sin^2 t + \cos^2 t = 1$$

$$\left(-\frac{1}{4}\right)^2 + \cos^2 t = 1$$

$$\cos t = -\frac{\sqrt{15}}{4} \quad \because \text{Q III}$$

$$\sec t = -\frac{4}{\sqrt{15}}$$

$$\sec t = -\frac{4\sqrt{15}}{15}$$

$$\csc t = -4$$

$$\tan t = \frac{-\frac{1}{4}}{-\frac{\sqrt{15}}{4}}$$

$$\tan t = \frac{1}{\sqrt{15}}$$

$$\tan t = \frac{\sqrt{15}}{15}$$

$$\cot t = \sqrt{15}$$

Example 7: Find functions f and g such that $F = f \circ g$.

a. $F(x) = \cos^2 x$

$$g(x) = \cos x$$

$$f(x) = x^2$$

b. $F(x) = \sqrt{1 + \tan x}$

$$g(x) = 1 + \tan x$$

$$f(x) = \sqrt{x}$$

Example 8: Determine whether the function is even, odd, or neither.

a. $f(x) = x^2 \cos(2x)$

$$f(-x) = x^2 \cos(-2x)$$

$$f(-x) = x^2 \cos(2x)$$

$$f(-x) = f(x)$$

even

b. $f(x) = x^3 + \cos x$

$$f(-x) = -x^3 + \cos(-x)$$

$$f(-x) = -x^3 + \cos(x)$$

$$f(-x) \neq f(x)$$

$$f(-x) \neq -f(x)$$

neither

Example 9: The displacement from equilibrium ($y = 0$) of an oscillating mass attached to a spring is given by $y(t) = 4\cos(3\pi t)$ where y is measured in inches and t in seconds. Find the displacement at the times indicated in the table.

t	$y(t)$
0	4 in
0.25	$-2\sqrt{2}$ in
0.50	0 in
0.75	$2\sqrt{2}$ in
1.00	-4 in
1.25	$2\sqrt{2}$ in

$$\begin{aligned} y(0) &= 4 \cos(0) \\ &= 4 \cdot 1 \\ &= 4 \end{aligned}$$

$$\begin{aligned} y(0.25) &= 4 \cos(3\pi \cdot 0.25) \\ &= 4 \cos\left(\frac{3\pi}{4}\right) \\ &= 4 \left(-\frac{\sqrt{2}}{2}\right) \\ &= -2\sqrt{2} \end{aligned}$$

$$\begin{aligned} y(0.50) &= 4 \cos(3\pi \cdot 0.5) \\ &= 4 \cos\left(\frac{3\pi}{2}\right) \\ &= 4 \cdot 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} y(0.75) &= 4 \cos(3\pi \cdot 0.75) \\ &= 4 \cos\left(\frac{9\pi}{4}\right) \\ &= 2\sqrt{2} \end{aligned}$$

$$\begin{aligned} y(1) &= 4 \cos(3\pi) \\ &= 4(-1) \\ &= -4 \end{aligned}$$

$$\begin{aligned} y(1.25) &= 4 \cos\left(\frac{15\pi}{4}\right) \\ &= 4\left(\frac{\sqrt{2}}{2}\right) \\ &= 2\sqrt{2} \end{aligned}$$

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Rebelle →

Don't Expose others